

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Ratios, rates, proportional relationships, and units

02

10 answers · Rationale-rich · Teach from doc

Q1

A

A. $\frac{1}{20}$

Choice A is correct. It's given that 3 feet = 1 yard. It follows that a speed of $\frac{3}{20}$ feet per second is equivalent to $\frac{\frac{3}{20} \text{ feet}}{1 \text{ second}} \frac{1 \text{ yard}}{3 \text{ feet}}$, which is equivalent to $\frac{3}{20} \frac{1}{3}$, or $\frac{1}{20}$, yards per second.

Choice B is incorrect. This is the speed, in feet per second, that's equivalent to $\frac{3}{20}$ yards per second.

Choice C is incorrect. This is the speed, in yards per second, that's equivalent to 18, not $\frac{3}{20}$, feet per second.

Choice D is incorrect. This is the speed, in yards per second, that's equivalent to 60, not $\frac{3}{20}$, feet per second.

Q2

73920

The correct answer is 73,920. It's given that 1 furlong = 220 yards and 1 yard = 3 feet. It follows that a distance of 112 furlongs is equivalent to $112 \text{ furlongs} \frac{220 \text{ yards}}{1 \text{ furlong}} \frac{3 \text{ feet}}{1 \text{ yard}}$, or 73,920 feet.

Q3**D****D. 2 : 1**

Choice D is correct. The cost per gram of protein in 1 large egg is $\$0.36 \div 6 = \0.06 . The cost per gram of protein in 1 cup of milk is $\$0.24 \div 8 = \0.03 . It follows that the ratio of the cost per gram of protein in a large egg to the cost per gram of protein in a cup of milk is 0.06:0.03, which can be rewritten as 2:1.

Choice A is incorrect and may result from finding the ratio of the cost per gram of protein in a cup of milk to the cost per gram of protein in a large egg (the reciprocal of the ratio specified in the question). Choices B and C are incorrect and may result from incorrectly calculating the unit rates or from errors made when simplifying the ratio.

Q4**980**

The correct answer is 980. It's given that the ratio 140 to m is equivalent to the ratio 4 to 28. Therefore, the value of m can be found by solving the equation $\frac{140}{m} = \frac{4}{28}$. Multiplying each side of this equation by m yields $140 = \frac{4m}{28}$. Multiplying each side of this equation by 28 yields $3,920 = 4m$. Dividing each side of this equation by 4 yields $980 = m$. Therefore, the value of m is 980.

Q5**370**

The correct answer is 370. It's given that the population density of Cedar County is 230 people per square mile and the county has a population of 85,100 people. Based on the population density, it follows that the area of Cedar County is $85,100 \text{ people} \frac{1 \text{ square mile}}{230 \text{ people}}$, or 370 square miles.

Q6**A**

A. 25 meters per second

Choice A is correct. Since 1 kilometer is equal to 1,000 meters, it follows that 90 kilometers is equal to $90(1,000) = 90,000$ meters. Since 1 hour is equal to 60 minutes and 1 minute is equal to 60 seconds, it follows that 1 hour is equal to $60(60) = 3,600$ seconds. Now $\frac{90 \text{ kilometers}}{1 \text{ hour}}$ is equal to $\frac{90,000 \text{ meters}}{3,600 \text{ seconds}}$, which reduces to $\frac{25 \text{ meters}}{1 \text{ second}}$ or 25 meters per second.

Choices B, C, and D are incorrect and may result from conceptual or calculation errors.

Q7**35728**

The correct answer is 35,728. It's given that $1 \text{ mile} = 1,760 \text{ yards}$. It follows that an average speed of 20.300 miles per hour is equivalent to $\frac{20.300 \text{ miles}}{1 \text{ hour}} \cdot \frac{1,760 \text{ yards}}{1 \text{ mile}}$, or 35,728 yards per hour.

Q8**4**

The correct answer is 4.44. The selling price, in dollars per cubic inch, is found by multiplying the density, in ounces per cubic inch, by the unit price, in dollars per ounce:

$\left(\frac{0.555 \text{ ounce}}{1 \text{ cubic inch}}\right)\left(\frac{\$8.00}{1 \text{ ounce}}\right)$ yields $\frac{\$4.44}{1 \text{ cubic inch}}$. Thus, the selling price, in dollars per cubic inch, is 4.44.

Q9**B****B. 8**

Choice B is correct. It is given that the volume of the glass is approximately 16 fluid ounces. If Jenny has 1 gallon of water, which is 128 fluid ounces, she could fill the glass $\frac{128}{16} = 8$ times.

Choice A is incorrect because Jenny would need 16×16 fluid ounces = 256 fluid ounces, or 2 gallons, of water to fill the glass 16 times. Choice C is incorrect because Jenny would need only 4×16 fluid ounces = 64 fluid ounces of water to fill the glass 4 times. Choice D is incorrect because Jenny would need only 3×16 fluid ounces = 48 fluid ounces to fill the glass 3 times.

Q10**16606**

The correct answer is 16,606. It's given that one side of a flat board has an area of 874 square inches. If a pressure of 19 pounds per square inch of area is exerted on this side of the board, the total force exerted on this side of the board is $874 \text{ square inches} \times \frac{19 \text{ pounds}}{1 \text{ square inch}}$, or 16,606 pounds.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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